EUI SCHOOL OF TRANSNATIONAL GOVERNANCE

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POLICY BRIEF

PRESERVING HUMAN AGENCY IN THE AGE OF AI

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EXECUTIVE SUMMARY

A mounting body of evidence shows artificial intelligence is fundamentally reshaping human experience, eroding essential capabilities, and threatening social cohesion. This policy brief synthesises key findings on the mounting crisis of human agency, documents concerning impacts already visible across society, and proposes a comprehensive framework for intervention. The evidence demonstrates a pattern of accelerating disconnection, skill erosion, and psychological strain that demands urgent action within a three- to five-year window before AI capabilities potentially reach levels that could permanently alter the human-technology relationship. Our approach emphasises the development of psycho-emotional resilience as the critical foundation for human flourishing in the AI age, complemented by supportive skill development and strengthened community structures.

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Views expressed in this publication reflect the opinion of individual authors and not those of the European University Institute.

1. INTRODUCTION

Current regulatory approaches, whilst essential, prove insufficient for addressing AI's cumulative impacts on human cognition, emotion, and social functioning. Like environmental contaminants that accumulate beneath detection thresholds, 'low-risk' AI systems collectively alter human agency through attention erosion, emotional dependency, and decision-making delegation. The youth mental health crisis correlates directly with smartphone adoption, whilst loneliness reaches epidemic proportions despite unprecedented connectivity.

This brief proposes psycho-emotional resilience as the foundation for human agency in the AI age, encompassing emotional control, flexible thinking, strong sense of self, confidence in action, and purpose and connection. These capabilities, supported by specific skills and community infrastructure, enable humans to thrive alongside AI rather than being diminished by it. Implementation requires coordinated action across families, educational institutions, workplaces, and civil society within the narrow window before AGI emergence potentially forecloses opportunities for meaningful intervention.

2. THE GATHERING STORM: TECHNOLOGY'S HIDDEN COSTS

2.1 The loneliness epidemic

Despite unprecedented technological connectivity, meaningful human connection is declining dramatically. Between 10 and 25% of Europeans report frequent loneliness, with particularly alarming rates among young adults (Schnepf et al, 2024). The US Surgeon General's 2023 advisory declared loneliness an epidemic, noting that social disconnection increases premature death risk by nearly 30% — comparable to smoking 15 cigarettes daily.

The most pronounced decline — a 70% reduction in social interaction — has occurred among young people aged 15-24 across advanced economies, suggesting that technology-driven approaches to connection may be perpetuating rather than solving the problem they claim to address. Digital substitution increasingly replaces real-world interactions with alternatives that provide the illusion of connection without its substance.

2.2 The youth mental health crisis

Depression rates among teenage girls doubled between 2010 and 2019, whilst emergency room visits for self-harm among girls aged 10 to 14 increased 189% from 2009 to 2019. The percentage of high school students reporting persistent sadness increased from 30% in 2013 to 40% in 2023 (Haidt, 2024).

Research by Jonathan Haidt (2024) identifies 2012 as a critical inflection point when smartphone adoption and social media reached critical mass. The aforementioned mental health trends consistently correlate with digital media exposure through mechanisms including sleep disruption, social comparison, fear of missing out, reduced face-to-face interaction, and cyberbullying extending into previously safe spaces.

2.3 Economic and physical costs

Despite massive digital investments, expected productivity gains have failed to materialise. Knowledge workers face mounting challenges from attention fragmentation, with Microsoft data showing that people participate in three times more meetings weekly than they did in 2020. The cumulative cost of digital distraction is estimated at 2 to 3% of GDP across developed economies — approximately €400 billion annually in the EU.

Physical impacts of increased digital technology use include substantial increases in neck disorders among office workers since 2010, rising myopia rates among European young adults, and significant sleep disruption, with average nightly sleep duration decreasing by 1.2 hours since 2000.

3. AI AS COGNITIVE ENVIRONMENTAL CONTAMINANT

3.1 The environmental analogy

When environmental scientists first measured trace heavy metals in water supplies, these levels were deemed 'acceptable' by regulatory standards. Yet decades later, the cumulative impacts of low-level heavy metal exposure have revealed devastating consequences. We now face a parallel situation with AI — not in our physical environment, but in our cognitive landscape.

Environmental health science highlights three critical patterns applying to AI exposure: cumulative effects exceeding individual impacts, latency between exposure and manifestation, and variable vulnerability across populations. Like lead exposure once considered safe at 'low levels,' we now recognise the possibility that no safe threshold exists for exposure to AI.

3.2 Pathways of cognitive impact

Regular interaction with systems that subtly shape cognitive and emotional processes without crossing perceived harm thresholds creates significant cumulative effects, particularly through the following mechanisms.

- **Attention erosion:** Each engagement-optimised interaction pushes the user's attention toward fragmentation. Daily exposure to digital distraction over the course of years reshapes neural pathways governing attention allocation, with measurable changes in attention span correlating with digital media consumption.
- **Emotional dependency:** Algorithms serving content optimised for emotional engagement create reward patterns that reshape emotional expectations. Users develop increasing dependency on algorithmically mediated stimulation, gradually eroding internal emotional regulation.
- **Decision-making independence:** Each delegated decision appears inconsequential, but collectively, such decisions reshape decision-making pathways. GPS dependency research demonstrates spatial navigation skill atrophy, with greater GPS use associated with poorer spatial memory during self-guided navigation (Maguire et al. 2000).

3.3 Regulatory gaps

Current AI governance assesses systems individually, missing possible combined impacts. This resembles early environmental regulation attempts that evaluated the impact of exposure to various chemicals in isolation. AI systems deemed 'low-risk' individually may collectively transform cognitive environments beyond resilience thresholds, creating 'cognitive externalities' that are currently unaccounted for in governance frameworks.

4. THE EROSION OF HUMAN CAPABILITIES

4.1 De-skilling and skill atrophy

AI drives unprecedented de-skilling across previously human-exclusive domains. Unlike earlier technological shifts affecting manual labour, AI targets cognitive functions from writing to decision-making. This creates concerning feedback loops: as technologies eliminate skill requirements (de-skilling), capabilities deteriorate through disuse (skill atrophy), increasing AI dependence and accelerating further deterioration.

Critical thinking gives way to AI-mediated decision-making, and information retention weakens as external systems become primary knowledge repositories. Emotional intelligence is also becoming a more vulnerable capacity as AI increasingly mediates social interactions.

4.2 Transformation of human connection

Dating apps dominate romantic partnerships, with over 50% of engaged couples having met through digital platforms as of 2025 (Crane, 2025). More concerning is the rise of AI companions designed to function as relationship partners. One such example, Replika, has attracted over 30 million users, with 85% reporting that they had developed

emotional connections with the AI companion. When romantic features were restricted in 2023, users experienced genuine 'breakup' distress.

AI also enables connections with digital representations of the deceased through 'digital ghosts' constructed from voices and text, potentially disrupting traditional grief processes. As AI becomes sophisticated at mimicking human interaction, boundaries between authentic and artificial connection blur, leaving people vulnerable to manipulation.

4.3 Threats to democratic functioning

These patterns collectively compromise the foundations of democratic governance. Democracy requires citizens to have accurate information, cognitive capacity for policy evaluation, institutional trust, willingness to engage with opposing viewpoints, and shared factual foundations — all increasingly undermined by technological trends.

Algorithmic filtering creates echo chambers isolating individuals within like-minded communities, contributing to affective polarisation whilst fragmenting shared understanding. The emergent 'post-truth' era is characterised by declining factual consensus and widespread difficulty with distinguishing authentic from manufactured information.

5. LIMITATIONS OF REGULATORY APPROACHES ALONE

Whilst frameworks like the EU AI Act provide essential guardrails, regulatory approaches alone prove insufficient. The traditional policymaking process often spans three to five years, whilst AI capabilities double every 18 to 24 months, creating inherent mismatches that grow more severe as AI systems become more advanced.

Current governance primarily addresses systems rather than individual experiences, struggling to capture cumulative psychological impacts of 'low-risk' systems on human agency. Regulations cannot fully address how technologies influence individual cognition and behaviour in daily life. Even perfect governance cannot eliminate all threats to agency, as advanced AI systems identify and exploit cognitive biases in ways that evade detection.

Skill atrophy occurs through individual usage patterns that policy cannot effectively regulate. When people consistently delegate cognitive functions to external systems, their capabilities diminish regardless of governance quality.

6. PSYCHO-EMOTIONAL RESILIENCE AS FOUNDATION

6.1 Defining mental resilience

The most fundamental response to the rise of AI involves developing psycho-emotional resilience — in other words, internal resources enabling individuals to maintain agency, meaning, and connection amid technological transformation. Unlike traditional resilience frameworks focused on external recovery, mental resilience addresses internal transformations needed for autonomy in an increasingly AI-mediated world.

Mental resilience encompasses five interconnected components directly addressing AI's challenges to human consciousness: emotional control, flexible thinking, strong sense of self, confidence in action, and purpose and connection. These capabilities can be intentionally developed, allowing individuals to shape their relationships with AI rather than being shaped by them.

6.2 Core components

- **Emotional control** involves recognising emotional triggers in algorithmic content and applying regulation strategies like mindfulness or cognitive reframing. This becomes essential as AI increasingly influences emotional environments through algorithmic content delivery.
- **Flexible thinking** encompasses adapting mindsets, seeing challenges as possibilities, and developing creative solutions when facing technological disruption. This skill reduces stress by turning technological obstacles into opportunities.

- **Strong sense of self** involves maintaining a clear identity that is based on personal values and remains coherent across online and offline contexts. This provides emotional anchoring when AI attempts to influence user cognition and behaviour through personalised content.
- **Confidence in action** means maintaining trust in one's own capabilities (despite the availability of AI alternatives) and taking initiative rather than defaulting to technological solutions. This enables people to take charge of their digital lives, rather than following AI's lead.
- **Purpose and connection** involves finding meaning through values-aligned goals and building supportive relationships beyond digital approximations. Such goals and relationships provide motivation and belonging that transcend technological mediation.

7. SKILLS FOR HUMAN FLOURISHING

7.1 Foundational skills

AI literacy involves understanding AI systems' capabilities, limitations, and biases at conceptual levels, enabling users to practise informed engagement rather than blind reliance. Information verification has become ever more crucial as AI-generated content proliferates, requiring users to develop skills in source triangulation and healthy scepticism toward unverified content.

Technological self-awareness — observing how technology influences thinking — acts as a crucial tool for maintaining agency. This metacognitive awareness helps individuals recognise when algorithmic filters function to narrow perspectives and aids in maintaining boundaries between judgement and technological influence.

7.2 Critical thinking and human connection

Independent analysis involves evaluating the underlying assumptions of AI output and applying integrative judgement across domains. Ethical Reasoning encompasses recognising the ethical dimensions of technological choices and participating in democratic technology governance.

Emotional Intelligence includes recognising authentic emotions and navigating complexity, whilst Human-AI Collaboration involves understanding the complementary strengths of humans and AI technology and maintaining human direction over technological systems.

8. THE ESSENTIAL ROLE OF COMMUNITIES

8.1 Community as a foundation for resilience

Whilst resilience manifests individually, it's fundamentally shaped by communal contexts. Families serve as primary resilience contexts through intergenerational transmission via emotional regulation modelling and identity provision. Educational institutions support resilience through critical digital literacy and social-emotional learning that can help address technological challenges.

8.2 Workplace and civil society

Workplace environments influence technological transition navigation through AI integration approaches. Organisations framing AI as enhancing rather than substituting human capabilities foster resilience by maintaining workers' sense of purpose. Community organisations provide contexts for developing meaning beyond technological metrics through civic engagement opportunities that channel personal agency into collective action.

9. IMPLEMENTATION FRAMEWORK

As AI systems become increasingly embedded in daily life, policymakers must act decisively to shape their societal impact through proactive intervention rather than reactive regulation. These recommendations provide a roadmap for building resilient, human-centered responses to technological transformation across immediate, medium-term, and long-term timeframes.

9.1 Immediate actions (2025-2026)

- **Educational integration:** Implement mental resilience education across curricula, integrate digital citizenship programmes addressing the psychological dimensions of technology, and establish service-learning projects applying technology to community challenges.
- **Workplace programmes:** Establish human-centred AI integration framing AI as augmentation, focus professional development on distinctly human skills, and create recognition systems celebrating irreplaceable human contributions.
- **Community infrastructure:** Create digital equity programmes addressing the access and psychological dimensions of AI, establish intergenerational programming, and develop community dialogue initiatives about technological impact.

9.2 Medium-term strategies (2026-2028)

- **Research and measurement:** Develop metrics for AI systems' cognitive, emotional, and social impacts; establish human agency indicator monitoring; and create synthetic population modelling for cumulative effect prediction.
- **Regulatory innovation:** Expand frameworks addressing cumulative 'low-risk' system impacts, mandate human-in-the-loop mechanisms for high-risk applications, and establish public AI utilities demonstrating ethical implementation.

9.3 Long-term vision (2028-2030)

- **Societal transformation:** Achieve widespread AI literacy as a prerequisite for democratic oversight, establish cultural norms prioritising human agency alongside technological efficiency, and create sustainable economic models ensuring broad AI benefit sharing.

10. CONCLUSION

The evidence demonstrates that the growth of AI has given rise to fundamental disruption patterns across multiple human functioning dimensions. The response window narrows as AI capabilities advance toward general intelligence (2027-2030) and potentially superintelligence (2030-2035).

This situation demands immediate attention due to rapidly accelerating AI development, young people's particular vulnerability during critical developmental periods, and democratic systems' fragility under multiple pressures. By combining robust regulatory approaches with targeted psycho-emotional resilience development, supportive skills, and community infrastructure, we can maintain human agency and wellbeing as AI capabilities advance.

The goal isn't to resist technological advancement but to ensure that AI serves human purposes, rather than reshaping humans for technological imperatives. Through intentional mental resilience development, complementary skills, and supportive communities, we can create futures where AI enhances rather than diminishes human flourishing.

The window remains open but requires immediate, sustained action across all societal levels. The choices made in coming years will likely determine whether AI becomes humanity's greatest tool for addressing global challenges or accelerates a transition to an era where human agency is gradually eroded through the silent accumulation of technological influence.

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